

Apache Airflow

- By Santhosh Kumar V S

Apache Airflow is an open-source workflow management tool used to automate tasks and workflows. It helps companies schedule tasks in the correct order and execute them automatically without manual work. Airflow is mainly used in data engineering, cloud computing, ETL processes, and report generation.

Airflow works like a smart scheduler. It automatically performs tasks at the required time and manages dependencies between tasks. Developers use Python language to create workflows in Airflow.

Many companies use Airflow because it reduces manual effort, saves time, and improves workflow management. It also helps monitor workflows and detect errors quickly.

Real-Time Example

An e-commerce company collects customer order details every hour. Airflow automatically collects data, cleans incorrect records, stores the data in the database, and generates reports.

Another example is Netflix. Netflix processes user viewing history, recommendation systems, and streaming reports using automated workflows.

Advantages of Airflow

Apache Airflow provides many advantages for organizations handling workflows and large amounts of data.

Easy Scheduling

Airflow can schedule tasks daily, weekly, monthly, hourly, or at custom times. This reduces manual work.

Apache Airflow

- By Santhosh Kumar V S

Automation

Airflow automates repetitive tasks and improves efficiency.

Monitoring

Users can monitor workflows using the Airflow dashboard and check task status easily.

Error Handling

If a task fails, Airflow can retry the task automatically without restarting the complete workflow.

Scalability

Airflow supports both small and enterprise-level applications.

Integration Support

Airflow supports AWS, Google Cloud, Hadoop, Spark, MySQL, PostgreSQL, and many other tools.

Open Source

Airflow is free to use and organizations can customize it according to their requirements.

Real-Time Example

A bank automatically generates customer transaction reports every night using Airflow.

Another example is a hospital that processes patient records daily and sends reports to management automatically.

Apache Airflow

- By Santhosh Kumar V S

Airflow Architecture

Apache Airflow architecture contains different components that work together to execute workflows efficiently.

Web Server

The web server provides the user interface where users can monitor workflows, check logs, and manage tasks.

Scheduler

The scheduler continuously checks workflows and decides when tasks should run.

Metadata Database

The metadata database stores workflow details, task history, execution logs, and user information.

Executor

The executor decides how tasks should run and sends tasks to workers.

Workers

Workers execute the tasks assigned by the executor.

DAG Folder

This folder contains Python files where workflows are created.

Workflow Process

1. Scheduler reads DAG files
2. Tasks are scheduled
3. Executor sends tasks to workers

Apache Airflow

- By Santhosh Kumar V S

4. Workers execute tasks

5. Results are stored in the database

Real-Time Example

A food delivery company processes customer orders, payment details, and delivery tracking using Airflow architecture.

DAG in Apache Airflow

DAG stands for Directed Acyclic Graph. It is the core concept in Apache Airflow.

Directed means tasks move in one direction.

Acyclic means tasks do not form loops.

Graph means tasks are connected together.

A DAG contains multiple tasks arranged in sequence based on dependencies.

Example Workflow

1. Extract Data
2. Clean Data
3. Transform Data
4. Store Data
5. Generate Reports

Each task starts only after the previous task is completed successfully.

Benefits of DAG

Apache Airflow

- By Santhosh Kumar V S

- Easy workflow management
- Better dependency handling
- Parallel task execution
- Easy monitoring and debugging

Real-Time Example

A university collects student attendance every day. Airflow first collects attendance data, validates records, updates the database, and sends reports to faculty members automatically.

Another example is weather forecasting systems where data collection, processing, prediction, and reporting are executed step by step.

Executors in Apache Airflow

Executors are responsible for running tasks in Airflow. Different executors are used based on project size and workload.

Sequential Executor

Sequential Executor runs one task at a time. It is mainly used for testing and learning purposes.

Advantages

- Simple configuration
- Easy to use
- Suitable for beginners

Real-Time Example

Apache Airflow

- By Santhosh Kumar V S

A student practicing Airflow on a personal laptop can use Sequential Executor.

Local Executor

Local Executor runs multiple tasks on the same machine.

Advantages

- Faster execution
- Better resource usage
- Parallel processing support

Real-Time Example

A small company generating employee reports daily can use Local Executor.

Celery Executor

Celery Executor distributes tasks across multiple worker systems.

Advantages

- Distributed task execution
- High scalability
- Faster processing

Real-Time Example

An e-commerce company processing millions of customer records can use Celery Executor.

Kubernetes Executor

Kubernetes Executor runs tasks inside Kubernetes pods.

Apache Airflow

- By Santhosh Kumar V S

Advantages

- Automatic scaling
- Efficient resource management
- Container-based execution

Real-Time Example

Cloud-based companies use Kubernetes Executor for scalable workflow execution.

Importance of Apache Airflow in Industry

Today many industries use Apache Airflow for workflow automation.

Banking Sector

Banks automate transaction processing and report generation using Airflow.

Healthcare Sector

Hospitals process patient records and medical reports automatically.

E-Commerce

Online shopping companies automate order tracking and sales analytics.

Education Sector

Universities automate attendance systems and result processing.

Apache Airflow

- By Santhosh Kumar V S

Entertainment Industry

Streaming platforms process user activity and recommendation systems.

Apache Airflow is one of the most useful workflow automation tools used in modern industries. It helps automate repetitive tasks, schedule workflows, manage dependencies, and monitor task execution efficiently.

Airflow architecture provides flexibility and scalability for both small and large applications. DAGs organize workflows clearly, while executors improve task execution performance.

Today Airflow is widely used in banking, healthcare, education, cloud computing, e-commerce, and big data industries. Learning Apache Airflow is very useful for students interested in data engineering, DevOps, cloud technologies, and workflow automation.